

Design of 2-Port Microstrip MIMO Antenna for Sub-6 GHz 5G Applications

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Abstract— This paper presents a compact two port Sub-6GHz MIMO antenna developed for emerging 5G wireless communication links. The proposed structure is fabricated on an FR-4 substrate ($\epsilon_r = 4.4$, thickness = 1.6 mm) with an overall size of 60×33 mm². The antenna operates from 3.9–6 GHz, covering major 5G bands including n77, n78, and n79. A modified crescent-shaped radiator is employed to enhance impedance bandwidth and radiation performance, while structural isolation techniques are incorporated to reduce mutual coupling between ports. Simulated results show good impedance matching with return loss below -10 dB, isolation greater than 21 dB, and VSWR below 2 (≈ 1.12) throughout the operating band. The antenna provides a peak realized gain of 3.9 dBi with stable radiation behavior. The minimal ECC value (≈ 0.0001) validates that the antenna elements operate with high degrees of independence. Given its miniaturized geometry and high isolation levels, the presented MIMO configuration is well-suited for next-generation 5G mobile communications.

Keywords— Gain, 5G (fifth-generation) Applications, Patch antenna, MIMO (multiple input multiple output), Envelope Correlation Coefficient (ECC)

I. INTRODUCTION

The fast expansion in wireless communication offerings along with the increasing quantity of connected gadgets have together greatly sped up the progress of 5G telecommunication networks. Features that need worldwide coverage splintering, large number of devices connected simultaneously, and the ability to operate over wide areas using little power include smart cities, Internet of Things (IoT), autonomous systems, and high-speed mobile broadband, all of which necessitate communication networks that are capable of supporting extremely fast data rates, minimal latency, and higher spectral efficiency. In order to fulfill these demands, 5G communication networks use millimeter waves and Sub-6GHz frequency bands. The Sub-6GHz frequency bands are more attractive in terms of coverage area and signal propagation properties [1].

Microstrip patch antennas receive considerable attention in wireless communication technologies because of advantages like low profile, compact size, ease of fabrication, and use of printed circuit technology. These features make them quite a good option for integration into the new generation of portable wireless devices. On the other hand, there are some limitations to conventional single-element microstrip antennas. This includes low gain, narrow bandwidth, and low spectral efficiency. Thus, these types of antennas are not suitable for the new 5G communication system [2].

To address these limitations, modern wireless systems increasingly employ MIMO antenna technology as an effective solution. In MIMO systems, both transmitter and receiver use multiple antenna elements, which helps increase link capacity, reliability, and throughput without requiring additional bandwidth or transmit power. [3]. However, implementing several antenna components within a limited space causes a significant issue called mutual coupling which is capable of impairing the performance of the system by leading to greater correlation between the antenna components. Thus, designing for very high isolation and very low correlation between MIMO components is a major goal in the development of miniaturized 5G antennas [4].

Various antenna structures and decoupling techniques have been put forward recently to cope with such challenges. For instance, several Sub-6GHz MIMO antennas have been designed using slot-based structures, monopole configurations and array arrangements to achieve improved bandwidth and radiation performance [5].

Other works have mixed concepts like the utilization of DGS, electromagnetic bandgap (EBG) structures, capacitive elements, neutralization lines, etc. to reduce the electromagnetic coupling between antenna elements [6].

Additionally, there are proposals for wideband MIMO antennas to allow multi-standard communications over the whole Sub-6GHz range, thus making them compatible with a number of wireless services including WLAN and 5G NR bands.

It is a difficult challenge to come up with a small, efficient MIMO antenna that has a large bandwidth, high isolation, and consistent far field behavior all at the same time. More specifically, antennas supporting multiple Sub-6GHz 5G bands like n77, n78, and n79 at the same time having compact size and good radiation efficiency is the topic of the ongoing research campaign [7].

In this work, we design a compact Sub-6GHz MIMO radiator for 5G wireless communication applications. The proposed MIMO radiator is fabricated on a FR-4 dielectric and operates over a broad frequency ranging from 3.9 to 6 GHz, thus covering all major frequency bands within the Sub-6GHz spectrum. A round patch is modified to generate crescent-shaped elements that not only improve the bandwidth of impedance but also the radiation of elemental characteristics of the antenna. In addition, the first method is the use of EBG structures, second is the modification of the ground plane with defects, third is the use of capacitive coupling elements and the fourth is the use of neutralization lines. These are the various isolation improvement techniques that we have applied to minimize electromagnetic coupling between antenna elements.

II. ANTENNA DESIGN

The proposed antenna is implemented on an FR-4 dielectric with relative permittivity ($\epsilon_r=4.4$), loss tangent 0.02, and thickness 1.6 mm. FR-4 substrates are widely used in practical antenna designs due to their low fabrication cost and mechanical stability. This antenna is designed for Sub-6GHz wireless communication systems, covering frequency ranging from 3.9 GHz to 6 GHz [7].

All electromagnetic simulations (reflection coefficient, operating bandwidth, radiation patterns) were carried out using ANSYS HFSS software.

The main process consists in first making a single radiating element and after that turning it into a MIMO structure. By first optimizing the single element, we get the right resonance features before putting together several elements.

A. Single Antenna Element Design

The shape of single radiating element is shown in Figure 1. Figure 1 shows the top and ground layer of single antenna element layout. The radiating patch is formed from a circular microstrip patch with a small portion of the circle removed to obtain a crescent shape to modify the surface current distribution pattern and effectively makes current path length longer, which contributes to better impedance matching and bandwidth improvement. In addition, a rectangular notch is cut close to the upper part of the patch to further tune the antenna behaviour. Recently, in Sub-6GHz antenna designs, a similar crescent-shaped radiating element has been used for achieving wideband performance [8].

Initial radius of the circular patch is calculated using the standard equation for a circular microstrip antenna:

$$r = \frac{F}{\sqrt{1 + \frac{2h}{\pi F \epsilon_r} \left[\ln\left(\frac{\pi F}{2h}\right) + 1.7726 \right]}}$$

$$F = \frac{8.791 \times 10^9}{f_r \sqrt{\epsilon_r}}$$

Here r = radius of the circular patch, f_r = resonant frequency, h = substrate thickness, ϵ_r = relative permittivity of substrate. The dimensions of antenna components derived from the analytical formulas were rechecked and improved by performing parametric simulations in HFSS. Table 1 presents the dimensions of single radiator element.

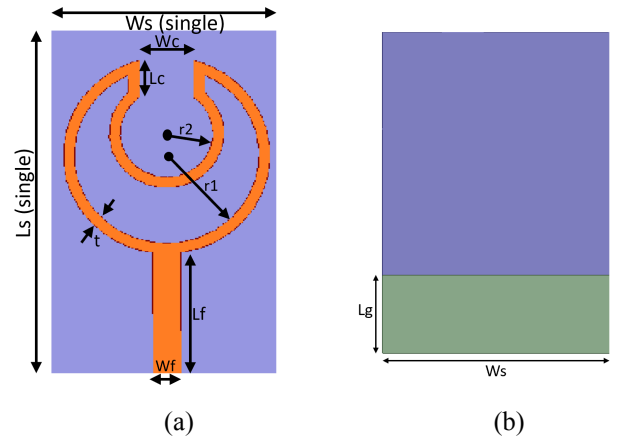


Figure 1: Structure of single-antenna element: (a) Top layer and (b) Ground layer.

Table 1: Single Antenna Dimensions

Antenna Parameters	Dimensions (mm)
$L_s(\text{single})$	33
$W_s(\text{single})$	22
L_f	11.655
L_c	3.5791
L_g	8
W_c	4.9
r_1	8.4
r_2	4.2
t	1.015
W_f	2.5

Figure 2 shows the S_{11} parameter or reflection coefficient of single-antenna element. From Figure 2, we observed that the antenna provides a good match for frequency ranging from 2.84 GHz to 6 GHz since the level of reflection coefficient is less than -10 dB. This corresponds to a broad operating bandwidth of nearly 4.2 GHz for the antenna. This bandwidth is wide enough to include very important

Sub-6GHz wireless frequency bands like 2.5GHz and 4.2GHz within the operating frequency band of radiator for the communication systems of wireless modern technology.

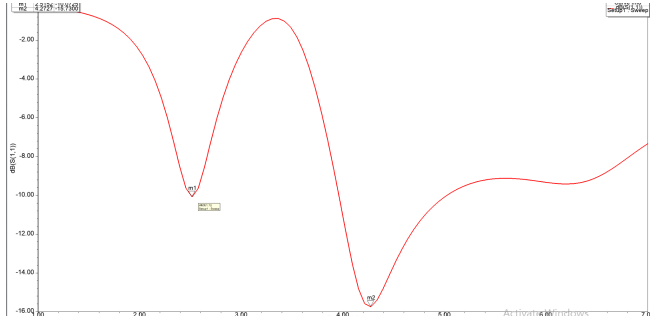


Figure 2: Reflection coefficient S_{11} of the individual antenna element.

Peak realized gain of simulated radiator is shown in Figure 3 below. The antenna design is such that the radiation characteristics remain constant over the operating range with the gain being above 2 dBi and reaching a maximum value of approximately 2.16 dBi at 4.2 GHz.

These results indicate that the proposed crescent-shaped radiating structure provides acceptable radiation performance for Sub-6GHz communication applications.

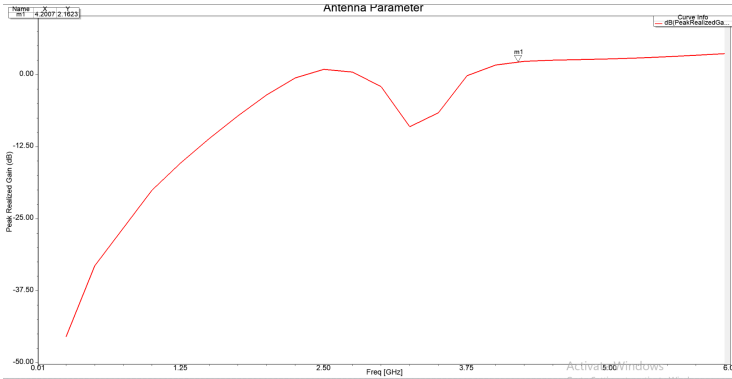


Figure 3: Peak realized gain of the single-antenna element.

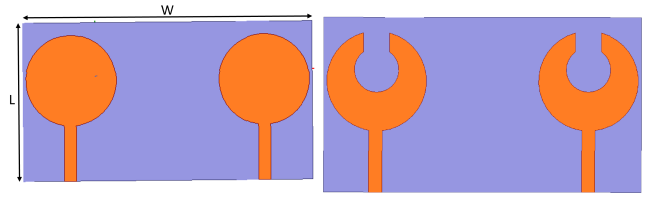
III. EVOLUTION STEPS

In step-1, a circular microstrip patch antenna was designed, which was fed by a microstrip transmission line on an FR-4 dielectric. To create a simple two-element structure, two identical circular radiating elements were symmetrically placed. This design gave the initial resonance but showed a narrow bandwidth and low performance.

In step-2, in order to enhance the features of our antenna, a circular slot was cut out inside each radiating patch so that the solid patch became a ring-like one. At the same time, a small opening was made at the upper part of the ring giving rise to a C-shaped radiating element. These changes disturbed the pattern of surface currents and lengthened the effective electrical path that led to changing the resonant frequency and better impedance matching [8].

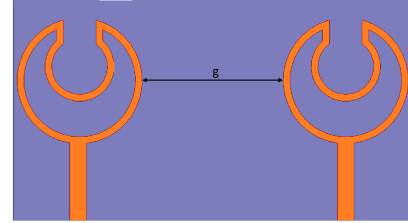
In step-3, the shape of the inner slot and the gap between the two antenna patches were fine-tuned. The fine-tuning

resulted in a concentric ring-type radiating structure, which had improved isolation and resonance characteristics, thus qualifying the antenna for Sub-6GHz MIMO applications.

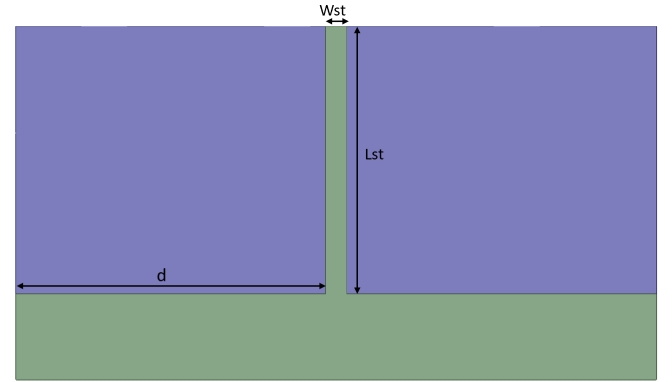


(a)

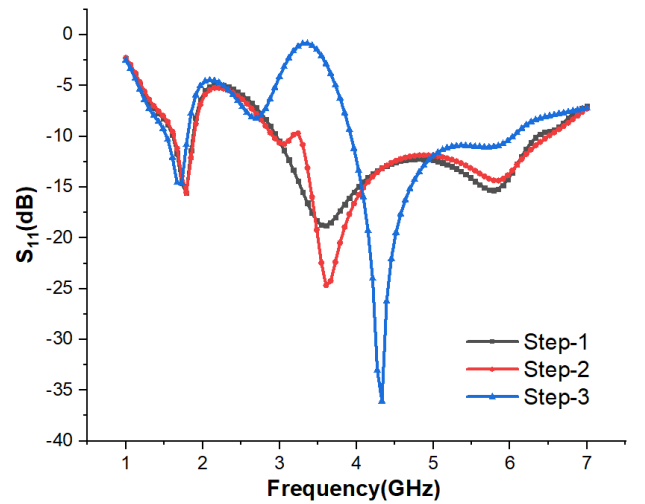
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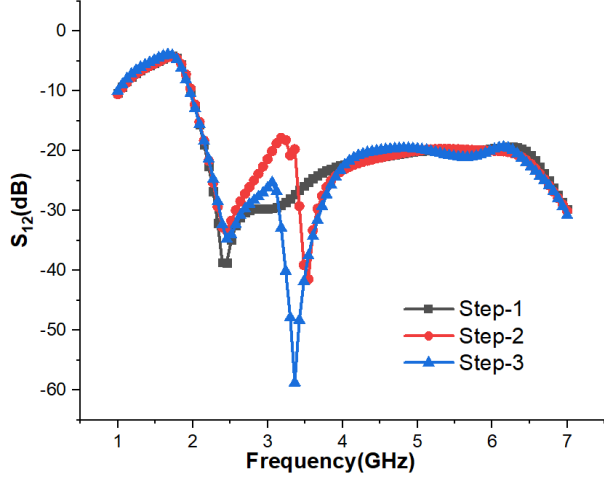
(c)



(d)



(e)



(f)

Figure 4: Antenna Evolution: (a) step 1, (b) step 2, (c) step 3 (proposed design), (d) back view, (e) S_{11} (steps), (f) S_{12} (steps)

Table 2: Proposed MIMO Antenna Dimensions

Antenna Parameters	Dimensions (mm)
W	60
L	33
g	21
Wst	2
Lst	25
d	29

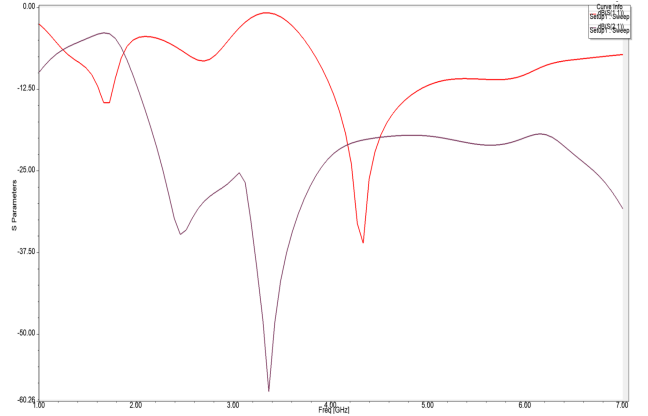
IV. RESULTS AND DISCUSSION

Simulated results validate the effectiveness of the proposed MIMO antenna. Strong impedance matching is achieved across the operating band, with a deep reflection minimum near 4.2 GHz reaching approximately -36 dB. The VSWR remains below 2, confirming efficient power transfer. Port isolation exceeds 21 dB, indicating effective suppression of mutual coupling. Far field analysis shows stable directional electromagnetic radiation behavior with a peak realized gain approximately equal to 4 dBi. E-plane and H-plane patterns exhibit broad main lobes without significant null distortion, supporting reliable radiation coverage. Furthermore, the envelope correlation coefficient remains nearly zero (≈ 0.0001), indicating excellent diversity characteristics.

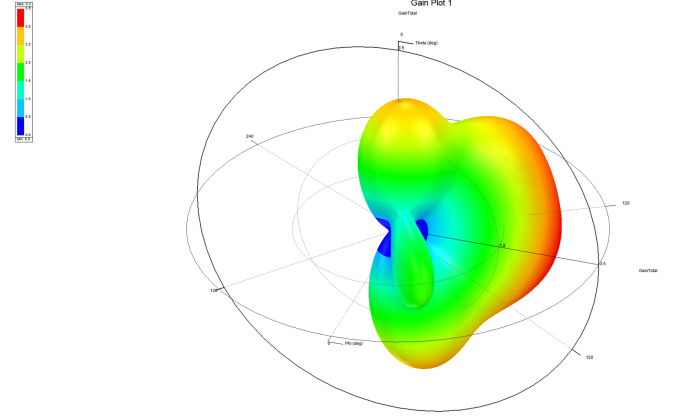
The antenna compared with recently published work is tabulated in Table 3 and has achieved lowest ECC, high frequency bandwidth and good isolation. The antenna is also highly compact as compared to previous related works

except [9](but is better in terms of ECC and frequency bandwidth).

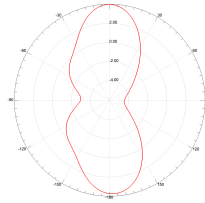
In particular, the proposed design demonstrates superior correlation performance while maintaining competitive bandwidth and reduced physical footprint.



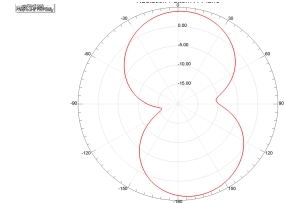
(a)



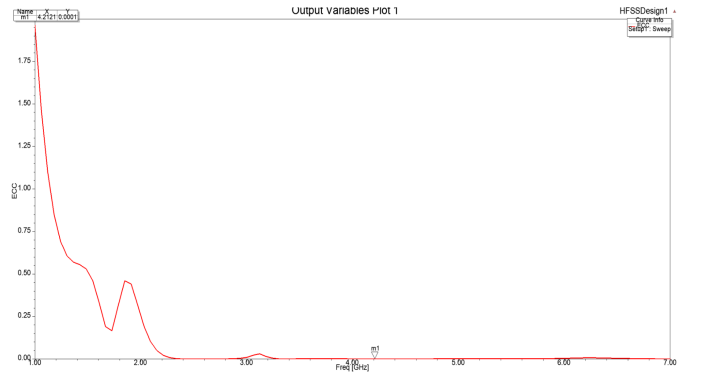
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(c)



(d)



(e)

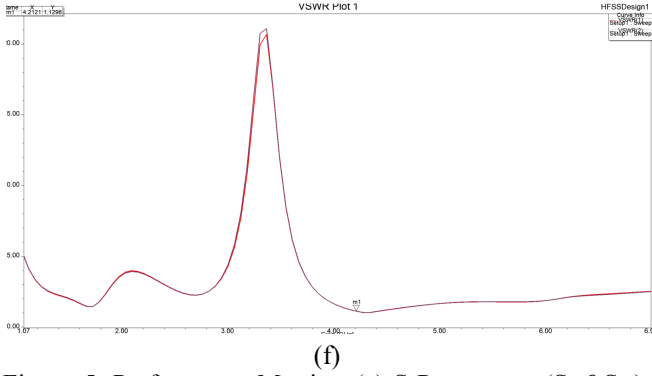


Figure 5: Performance Metrics: (a) S-Parameters (S_{11} & S_{12}), (b) 3D Gain Plot, (c) E-Plane and (d) H-Plane polar plot, (e) ECC, (f) VSWR.

Table 3: Proposed Antenna As Compared to Related Published Work

Ref no.	Area	Freq. band	Gain	Isolation (dB)	ECC
[8]	140x13 5	3.5 5.8	7.66 7.84	-	0.02
[9]	40x40	3.2 5.75	9	>22	<0.05
[10]	160x70	5.6-5 .67	12.4	>30	0.02
[11]	60x60	3.4-3 .8	4.5	>19	<0.12
[12]	73.5x4 2	3.4-3 .6	5.4	>16	<0.03
[13]	120x60	3.33- 5.66	-	-	<0.26
Proposed Work	60x33	3.9-6	3.9	>21	0.0001

V. CONCLUSION

A space-efficient, two port MIMO radiator designed for Sub-6GHz 5G applications is proposed in this paper. The proposed design achieves broad operating bandwidth from 3.9–6 GHz together with strong port isolation (>21 dB), low correlation, and stable radiation behavior within a compact $60 \times 33 \text{ mm}^2$ footprint. Compared with related reported designs, the antenna offers an effective tradeoff between size, bandwidth, isolation, and diversity performance, with

particularly strong ECC performance ($\text{ECC} \approx 0.0001$). These characteristics make it a promising candidate for compact handheld 5G terminals. Future work will focus on fabrication and measurement-based validation of the proposed design.

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